Enthalpy method for the Stefan problem and a heuristic extension to Cattaneo-type conduction

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These notes describe the schemes implemented in the GitHub repository `Stefan-Problem-Enthalpy-Method`, the references on which they are based, and the tests used to validate them. The first part deals with the classical (Fickian) Stefan problem, for which the enthalpy method is standard and well understood. The second part is a numerical exploration of the case in which Fourier's law is replaced by the Cattaneo law. In this setting there is no well-established convergence theory, so this part should be read as heuristic.

1 The problem

We consider a material with two phases, solid and liquid, with melting temperature T_m. In each phase the temperature satisfies the heat equation, $\rho c_s T_t = k_s \Delta T$ in the solid and $\rho c_l T_t = k_l \Delta T$ in the liquid, and on the interface $\Gamma(t)$ we have $T = T_m$ and the Stefan condition

$$\rho \mathcal{L}\, V = \big[-k \nabla T \cdot \nu \big]_s^l, \tag{1}$$

where $\mathcal{L}$ is the latent heat, ν is the unit normal to Γ pointing towards the liquid, V is the normal velocity of the front in the direction of ν, and $[\,\cdot\,]_s^l$ denotes the value on the liquid side minus the value on the solid side. The density ρ is assumed to be the same in both phases.

We take a length scale ℓ, a temperature scale ΔT, and time in units of ℓ^2/α_l, with $\alpha_l = k_l/(\rho c_l)$. Writing $\theta = (T - T_m)/\Delta T$, the problem depends only on the Stefan number $\mathrm{St} = c_l \Delta T/\mathcal{L}$ and on the ratios $r_c = c_s/c_l$ and $r_k = k_s/k_l$. The enthalpy (sensible plus latent heat, in units of $\rho c_l \Delta T$) and the Kirchhoff potential are

$$H = \begin{cases} r_c\, \theta, & \theta < 0, \\ \theta + 1/\mathrm{St}, & \theta > 0, \end{cases} \qquad u = \begin{cases} r_k\, \theta, & \theta < 0, \\ \theta, & \theta > 0, \end{cases}$$

and at $\theta = 0$ the enthalpy takes any value in $[0, H_m]$, $H_m = 1/\mathrm{St}$. In terms of these variables the whole problem, including (1), is the single equation

$$H_t = \Delta \beta(H), \qquad \beta(H) = \alpha_s \min(H, 0) + \max(H - H_m, 0), \tag{2}$$

understood in the weak sense, where $u = \beta(H)$ and $\alpha_s = r_k/r_c$ is the diffusivity of the solid relative to the liquid. The solid is $\{H < 0\}$, the liquid is $\{H > H_m\}$, and the set $\{0 \le H \le H_m\}$ is at melting temperature. The front is never computed explicitly; it is recovered as the level set $\{f = 1/2\}$ of the liquid fraction $f = \min(1, \max(H/H_m, 0))$, which allows fronts to merge or disappear without any special treatment. This formulation is classical; a standard reference for the method is [1].

Equation (2) is a degenerate parabolic equation with β nondecreasing and Lipschitz, with constant $L_\beta = \max(1, \alpha_s)$. For this class of equations the operator $H \mapsto -\Delta\beta(H)$ is m-accretive in L^1 [3], so by the Crandall–Liggett theorem [6] it generates a semigroup of contractions in L^1, which is also order preserving. The properties of the scheme below are discrete counterparts of this structure.

2 The enthalpy scheme

We use a uniform mesh of cubic cells of size h on a box in $\mathbb{R}^d$, $d = 1, 2, 3$. The unknown H_i^n is the average of H on the cell i at time t_n, and $U_i^n = \beta(H_i^n)$. The scheme is explicit,

$$H_i^{n+1} = H_i^n + \Delta t\,(\Delta_h U^n)_i, \qquad (\Delta_h U)_i = \frac{1}{h^2}\sum_{j\sim i}(U_j - U_i), \tag{3}$$

where $j \sim i$ runs over the $2d$ neighbours of i, so Δ_h is the usual $(2d+1)$-point Laplacian. This is the explicit enthalpy method as described in [1]; it can also be seen as a finite volume scheme in which the flux through each face is $-(U_j - U_i)/h$.

The boundary conditions are imposed with a layer of ghost cells. For a Dirichlet condition $\theta = \theta_b$ on a face we set the ghost value $U_g = 2u_b - U_a$, where U_a is the adjacent inner cell and u_b is the Kirchhoff potential of θ_b, so that the mean of both is u_b on the face. For a Neumann condition with incoming flux g we set $U_g = U_a + hg$. The data θ_b and g may depend on time and on the position along the face.

Since $0 \le \beta' \le L_\beta$, the right-hand side of (3) is nondecreasing in every H_j^n provided that

$$\Delta t \le \frac{h^2}{L_\beta\,\mathrm{diag}}, \tag{4}$$

where $\mathrm{diag} = 2d + m$ and m is the number of axes with a Dirichlet face (a Dirichlet face adds 1 to the diagonal coefficient of the adjacent cell). This is the usual monotonicity condition for explicit schemes for (2), see [1, 9]. In the code we take Δt equal to 0.9 times this bound. Under (4) the scheme is monotone and therefore satisfies a discrete comparison principle. Since it is also written in conservative form, the Crandall–Tartar lemma [7] shows that it is an L^1 contraction (for no-flux or periodic boundary conditions). These are the discrete versions of the properties of the semigroup, and convergence results for discretizations of (2) rely on them, see for instance [2, 9].

The scheme is also exactly conservative. If the heat that enters through the boundary faces is accumulated at each step, the total enthalpy $h^d \sum_i H_i^n$ changes exactly by that amount, which we use as a test of the implementation.

3 Tests for the Fickian case

The first test is the Neumann solution [4, 1]. On $x > 0$ the solid is initially at $\theta = -\theta_s$, and $\theta = 1$ is prescribed at $x = 0$. The front is $s(t) = 2\lambda\sqrt{t}$, where λ is the root of a transcendental equation coming from (1), and the temperature is given by error functions in each phase. We take $\mathrm{St} = 1$, $r_c = 1$, $r_k = 2$ and $\theta_s = 1/2$, which gives $\lambda \approx 0.4420$. We solve on $[0, 8]$ with an insulated wall at $x = 8$ (far enough to approximate the half line), starting at $t_0 = 0.1$ from the cell averages of the exact enthalpy, and compare at $T = 1$. It is important to average H rather than θ; otherwise the cell cut by the front loses part of its latent heat and the error of the front stops decreasing.

N	L^1 error in θ	L^∞ error in θ	front error	energy balance
200	2.78×10^{-3}	1.15×10^{-2}	2.52×10^{-3}	2.2×10^{-15}
400	1.07×10^{-3}	5.37×10^{-3}	1.25×10^{-3}	6.7×10^{-16}
800	3.53×10^{-4}	1.22×10^{-3}	4.76×10^{-4}	1.2×10^{-14}
1600	1.46×10^{-4}	1.47×10^{-3}	4.26×10^{-5}	1.1×10^{-13}

Table 1: Neumann solution at $T = 1$ with N cells on $[0, 8]$. The front error compares the liquid volume with $s(T)$, and the energy balance is the change of total enthalpy minus the heat that entered through the boundary.

The L^1 error and the error of the front decrease at least like $O(h)$. The L^∞ error decreases more slowly, since it is attained next to the front, where the explicit scheme produces the small plateaus typical of the enthalpy method (the temperature of a cell remains at zero while it absorbs the latent heat). The energy balance holds up to rounding error. We also checked the energy balance in 2d and 3d with random initial data and mixed Dirichlet and Neumann conditions, obtaining again values of order 10^{-15}.

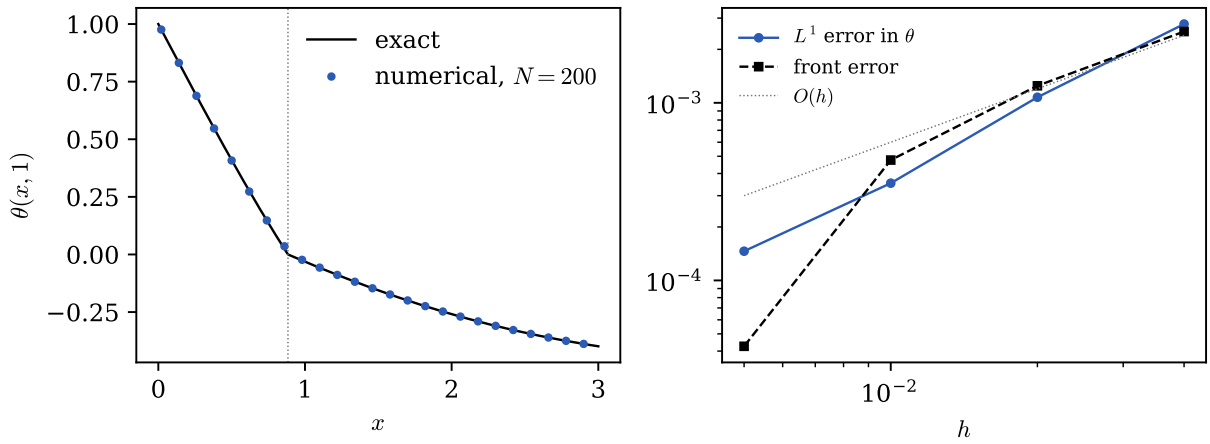


Figure 1: Left, temperature at $T = 1$ for $N = 200$ against the Neumann solution (the dotted line is the exact front). Right, L^1 error in θ and error of the front as functions of h.

The remaining examples in the repository are mainly visualizations with more complex,

dendritic initial data. In 1d we take random layers of ice and water between a hot and a cold wall, in 2d a snowflake, and in 3d a cubic dendrite, the last two colder than the surrounding water. Figure 2 shows some snapshots of the 2d case, where the bottom wall is cold and the top wall is warm. The water between the branches freezes first and the tips melt, so the crystal becomes rounder before it merges with the ice growing from the bottom.

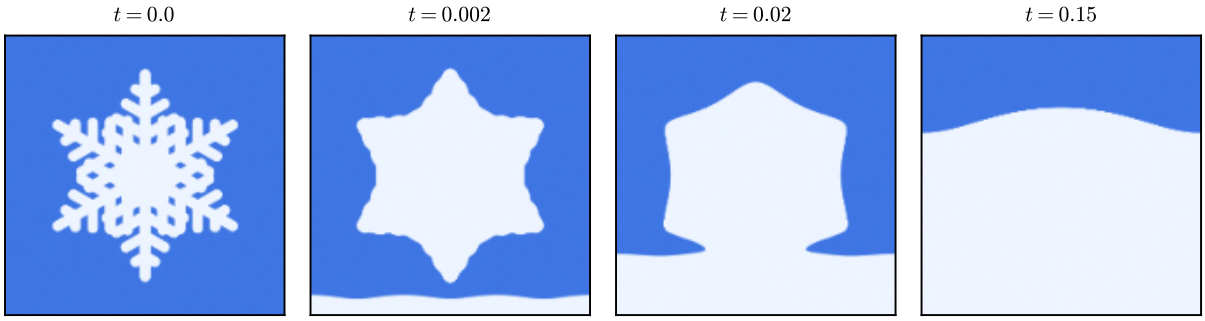


Figure 2: Liquid fraction for the 2d snowflake (St $= 0.5$, $r_k = 2$, $N = 200$). The solid is light and the liquid is blue.

4 A formal extension to Cattaneo-type conduction

We now consider the case in which Fourier's law is replaced by the Cattaneo law [5], a relaxation equation for the heat flux, $\tau q_t + q = -k\nabla T$, with relaxation time $\tau > 0$. After the same scaling as before (with τ in units of ℓ^2/α_l), and since $k\nabla\theta = \nabla u$ inside each phase, the natural enthalpy form of the problem is

$$H_t + \nabla \cdot q = 0, \qquad \tau q_t + q = -\nabla u, \qquad u = \beta(H). \tag{5}$$

Eliminating q formally gives $\tau H_{tt} + H_t = \Delta\beta(H)$, a damped and degenerate wave equation, with local characteristic speeds $\pm\sqrt{\beta'(H)/\tau}$, that is, $1/\sqrt{\tau}$ in the liquid, $\sqrt{\alpha_s/\tau}$ in the solid and 0 at melting temperature. For $\tau = 0$ we recover (2).

We emphasize that this extension is only formal. The theory of nonlinear semigroups underlying the Fickian scheme does not extend directly to this setting. The operator associated with the system in the variables (H, q) is not accretive in L^1 in any obvious way, and the telegraph equation does not satisfy a comparison principle, so a convergent scheme for (5) cannot be monotone. Hence the L^1 contraction and the comparison principle are lost, and with them the standard route to convergence. Well-posedness results for hyperbolic versions of the Stefan problem exist [12, 8], but they rely on other methods, and different ways of coupling the Cattaneo law with the phase change lead to problems that coincide as $\tau \to 0$ but not for $\tau > 0$. In what follows we simply discretize (5) and test the result against an exact solution.

Since both equations in (5) are in divergence form (the term q is a bounded source), across a front with normal velocity V the Rankine–Hugoniot conditions

$$V[H] = [q \cdot \nu], \qquad \tau V[q \cdot \nu] = [u] \tag{6}$$

must hold, and together they give $[u] = \tau V^2 [H]$. For $\tau = 0$ this reduces to $[u] = 0$, so the temperature is continuous and equal to the melting temperature on the front. For $\tau > 0$ and a moving front, instead, the temperature must jump. This is the main new feature of the problem, and also the source of the numerical difficulties discussed below.

5 A travelling wave

To test the scheme we look for travelling wave solutions of (5) in 1d, $H = H(\xi)$, $q = Q(\xi)$ with $\xi = x - s(t)$ and $s(t) = s_0 + Vt$. Ahead of the front ($\xi > 0$) we place solid at melting temperature, $H = u = Q = 0$, and behind it ($\xi < 0$) liquid, where $u = H - 1/\mathrm{St}$. In the liquid the first equation gives $-VH' + Q' = 0$, so $Q = Vu + c$ for some constant c, and substituting in the second equation, $-\tau V Q' + Q = -u'$, we get

$$(1 - \tau V^2)\, u' = -Vu - c.$$

We write $b = 1 - \tau V^2$. Then $u = -c/V + Ae^{-V\xi/b}$, and for u to grow towards the hot side ($\xi \to -\infty$) we need $b > 0$, that is, $V < 1/\sqrt{\tau}$; the front moves slower than the thermal waves in the liquid. Since the state ahead of the front is zero, conditions (6) read $Q^- = V(u^- + 1/\mathrm{St})$ and $u^- = \tau V Q^-$, where the superscript denotes the value on the liquid side. Solving, $u^- = \tau V^2/(\mathrm{St}\, b)$, $Q^- = V/(\mathrm{St}\, b)$, $c = V/\mathrm{St}$ and $A = 1/(\mathrm{St}\, b)$. Hence

$$u = \frac{1}{\mathrm{St}}\Big(\frac{e^{-V\xi/b}}{b} - 1\Big), \qquad q = \frac{V}{\mathrm{St}\, b}\, e^{-V\xi/b} \qquad (\xi < 0), \qquad u = q = H = 0 \qquad (\xi > 0). \quad (7)$$

For $\tau = 0$ this is the classical wave with $u = 0$ on the front, while for $\tau > 0$ the temperature on the liquid side of the front is $u^- = \tau V^2/(\mathrm{St}\, b) > 0$. We remark that, as for any hyperbolic system, conditions (6) do not select admissible solutions by themselves. If the solid ahead of the front is below the melting temperature, there are two families of waves satisfying (6), one with the solid at melting temperature on the front (as above) and one with the liquid at melting temperature and an undercooled solid. In our tests the numerical solutions follow only the first one. Which of them is physically relevant is a question of admissibility that we do not address here.

6 Schemes for the Cattaneo case

The natural discretization of (5) keeps H_i on the cells, as before, and puts each component q_a of the flux on the faces normal to the axis a, which is a staggered mesh. We denote by ∇_h the difference from cells to faces, $(\nabla_h U)_{i+e_a/2} = (U_{i+e_a} - U_i)/h$, and by $\nabla_h \cdot$ the difference from faces to cells, so that $\nabla_h \cdot \nabla_h = \Delta_h$. Treating the relaxation term implicitly and the transport explicitly, we obtain the first scheme we tested,

$$q^{n+1} = \frac{\tau q^n - \Delta t\, \nabla_h U^n}{\tau + \Delta t}, \qquad H^{n+1} = H^n - \Delta t\, \nabla_h \cdot q^{n+1}. \quad (8)$$

For $\tau = 0$ the first equation gives $q^{n+1} = -\nabla_h U^n$, so (8) is exactly (3), and it is conservative in H by construction. It reuses all the components of the Fickian solver (ghost cells, boundary conditions, β); only the time step and the update change. The boundary conditions enter

through $\nabla_h U$ on the boundary faces, and the incoming heat is now computed with q on those faces.

Scheme (8) is stable and conservative, but it does not converge to the travelling wave (7). For $\tau > 0$ it produces oscillations behind the front, at the scale of the mesh, which do not decrease as the mesh is refined (Figure 3). Heuristically, (8) is a centered scheme for a hyperbolic system and has no numerical dissipation. The front carries a jump in u, and each time it crosses a cell it emits short waves that travel backwards at speed $1/\sqrt{\tau}$ and are damped by the relaxation only at a rate of order $1/\tau$. For small τ (in our test, $V\sqrt{\tau} = 0.1$) the effect is small, but for $V\sqrt{\tau} \gtrsim 0.3$ it dominates the solution.

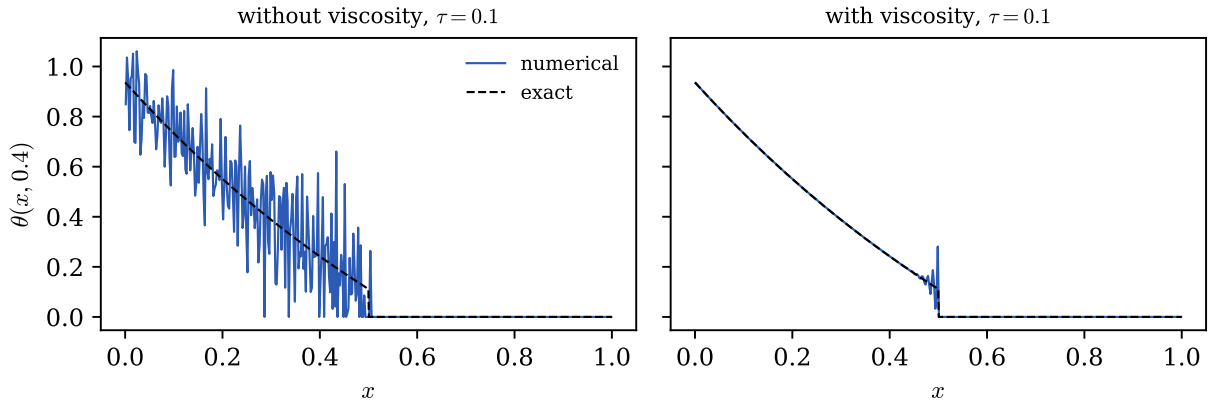


Figure 3: Travelling wave with $\tau = 0.1$ at $t = 0.4$, $N = 400$. Left, scheme (8). Right, scheme (9) with $\kappa = 0.25$.

The simplest remedy is to add a small numerical viscosity. The scheme finally implemented is

$$q^{n+1} = \frac{\tau q^n - \Delta t\, \nabla_h U^n}{\tau + \Delta t}, \qquad H^{n+1} = H^n - \Delta t\, \nabla_h \cdot q^{n+1} + \Delta t\, \varepsilon\, \Delta_h U^n, \tag{9}$$

with $\varepsilon = \kappa\, h \sqrt{L_\beta/\tau}$ for $\tau > 0$ and $\varepsilon = 0$ for $\tau = 0$. The extra term is ε times the step of the Fickian scheme. Its size is h times the largest characteristic speed, as in the numerical viscosity of Lax–Friedrichs type schemes [11], so it is consistent and vanishes as $h \to 0$. We take $\kappa = 0.25$, which was enough to remove the oscillations in our tests. The drawback is that ε grows as $\tau \to 0$ for fixed h, so (9) does not behave uniformly in τ. Capturing the jump while keeping the correct parabolic limit is precisely the problem of designing asymptotic preserving schemes [10], which we do not pursue here.

Linear stability. The time step is chosen from a von Neumann analysis of the linearized scheme, with $\beta(H) = LH$, on a periodic mesh. Let $k^2 = \sum_a 4 \sin^2(\xi_a h/2)/h^2 \in [0, 4d/h^2]$ be the symbol of $-\Delta_h$ for a wave number ξ, and write $\mu = Lk^2$, $a = \tau/(\tau + \Delta t)$ and $b = \Delta t/(\tau + \Delta t)$. Taking the Fourier mode of H and of $p = \nabla_h \cdot q$, scheme (9) becomes

$$\begin{pmatrix} H \\ p \end{pmatrix}^{n+1} = \begin{pmatrix} 1 - \Delta t\,(b + \varepsilon)\mu & -\Delta t\, a \\ b\mu & a \end{pmatrix} \begin{pmatrix} H \\ p \end{pmatrix}^{n},$$

(the components of q orthogonal to ξ are only multiplied by a). The determinant of this matrix is $a(1-\Delta t\,\varepsilon\mu)$ and its trace is $1+a-\Delta t\,(b+\varepsilon)\mu$. By the Schur–Cohn criterion both eigenvalues are in the open unit disk if and only if the determinant has modulus less than 1 and the trace has modulus less than 1 plus the determinant. The upper inequality on the trace reduces to $b(1+\varepsilon)\mu > 0$, which always holds for $k \neq 0$. The lower one reads $\Delta t\,\mu\,(b+\varepsilon+a\varepsilon) < 2(1+a)$, and multiplying by $\tau+\Delta t$ it becomes

$$(1+\varepsilon)\,\mu\,\Delta t^2 + 2(\varepsilon\mu\tau - 1)\,\Delta t - 4\tau < 0. \tag{10}$$

It also implies $\Delta t\,\varepsilon\mu < 2$, so the condition on the determinant holds. The worst case is $\mu = 4dL/h^2$. In the code we use $L = L_\beta$ and, as in (4), replace $2d$ by diag to account for the Dirichlet faces, so $\mu = 2L_\beta\,\mathrm{diag}/h^2$, and we take 0.9 times the positive root of (10). For $\tau = 0$ ($\varepsilon = 0$) it gives exactly (4), and for $\varepsilon = 0$ and $\tau \gg \Delta t$ it gives $\Delta t \lesssim h\sqrt{\tau/(L_\beta d)}$, a hyperbolic CFL condition. This is only a linear analysis, and for $\tau > 0$ neither scheme is monotone.

7 Tests for the Cattaneo case

We solve on $[0,1]$ with $\mathrm{St} = 1$, $r_c = r_k = 1$ and $V = 1$, with the front moving from $s_0 = 0.1$ to 0.5 ($T = 0.4$). We impose the exact temperature at $x = 0$ and $\theta = 0$ at $x = 1$, and start from the cell averages of the exact enthalpy and the exact flux $q = VH$ on the faces. We first checked that for $\tau = 0$ scheme (9) gives the same result as the Fickian solver up to rounding (differences of order 10^{-16}, with the same time step). Table 2 compares both schemes for $\tau = 0.01, 0.1, 0.5$. In all cases the energy balance holds up to rounding (at most 3×10^{-15}).

		scheme (8)		scheme (9), $\kappa = 0.25$	
τ	N	L^1 error in H	front error	L^1 error in H	front error
0.01	100	3.57×10^{-3}	$< 10^{-5}$	7.92×10^{-3}	5.95×10^{-3}
	800	3.80×10^{-3}	1.40×10^{-5}	1.26×10^{-3}	7.52×10^{-4}
0.1	100	6.37×10^{-2}	1.96×10^{-3}	5.85×10^{-3}	6.94×10^{-4}
	800	9.09×10^{-2}	1.16×10^{-4}	7.98×10^{-4}	8.82×10^{-5}
0.5	100	1.43×10^{-1}	$< 10^{-5}$	1.01×10^{-2}	1.89×10^{-3}
	800	1.25	6.47×10^{-2}	1.55×10^{-3}	1.71×10^{-4}

Table 2: Travelling wave at $t = 0.4$ with N cells on $[0,1]$. The front error compares the liquid volume with $s(t)$. For $\tau = 0$ both schemes coincide with the Fickian one, with L^1 error 2.75×10^{-4} for $N = 100$ and 1.52×10^{-5} for $N = 800$.

Without viscosity, the L^1 error in H does not decrease with N for any $\tau > 0$, and for $\tau = 0.5$ it grows. For $\tau = 0.01$ the front is placed correctly, but the error in H remains around 3.6×10^{-3}. With the viscosity the errors decrease at first order for every τ (Figure 4). The case $\tau = 0.01$ illustrates the drawback mentioned above; there ε is relatively large, the viscosity dominates the error, and on coarse meshes the error is larger than without viscosity. A small oscillation of a few cells remains near the front, located at the jump of the temperature (Figure 3, right). The left panel of Figure 4 also shows the effect of the

relaxation on the wave for a fixed speed V. As τ increases, the profile in the liquid becomes steeper and the jump of the temperature at the front grows, from 0 for $\tau = 0$ to 1 for $\tau = 0.5$.

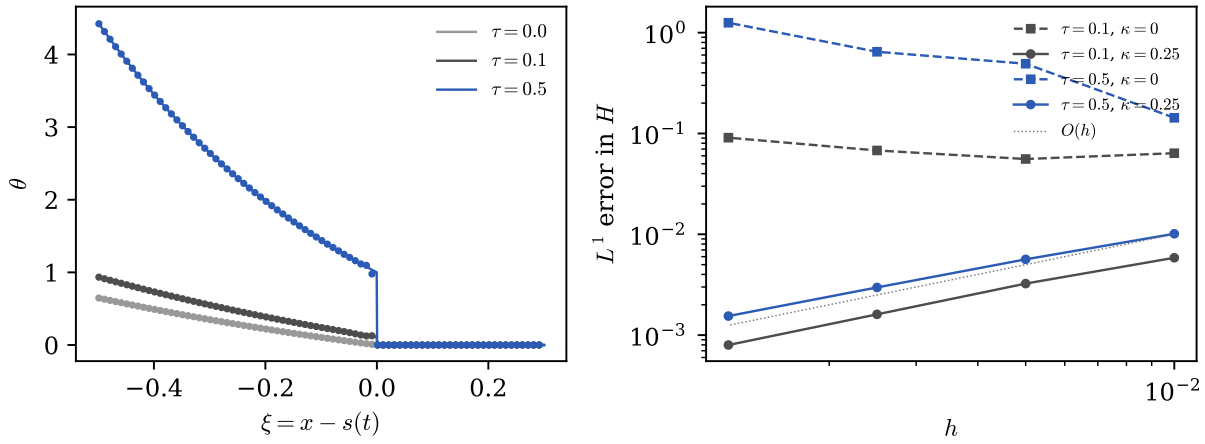


Figure 4: Left, exact travelling waves (lines) and numerical solutions of (9) with $N = 400$ (dots) in the moving frame. Right, L^1 error in H as a function of h, for scheme (8) ($\kappa = 0$, dashed) and scheme (9) ($\kappa = 0.25$, solid).

In summary, in the Fickian case the explicit enthalpy method behaves as expected from the theory. In the Cattaneo case the direct extension of the scheme is stable and conservative but does not converge, at least to the travelling wave (7). A numerical viscosity of order h recovers first order convergence in our tests, at the cost of losing the uniform behaviour as $\tau \to 0$. We do not have a proof of convergence for (9), and the question of which weak solutions of (5) are admissible is left open.

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